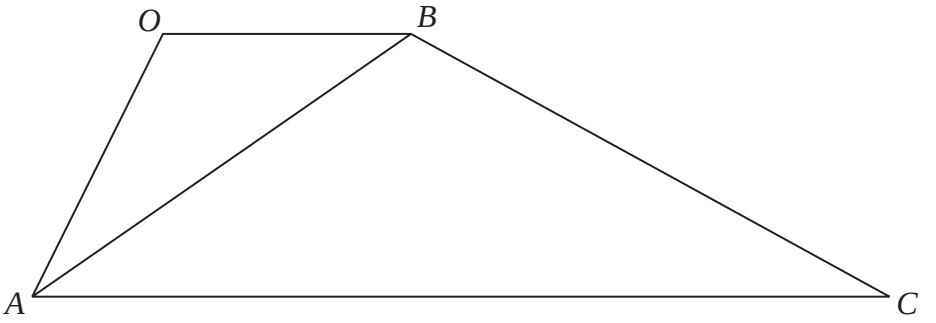
20



The diagram shows a trapezium $AOBC$.
AC is parallel to OB.
$AC = 4OB$.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

The point P lies on the line AB such that $AP : PB = 4 : 1$

Using a vector method, prove that OPC is a straight line.

All named expressions below are vectors.

$AB = OB - OA = b - a$

1 mark
Find AB.

$AP = (4/5)AB = (4/5)(b - a)$

1 mark
AP is4 of5 parts.

$OP = OA + AP = a + (4/5)(b - a)$

$= (1/5)a + (4/5)b$

1 mark
Complete routes from the same point O.

$OC = OA + AC = a + 4b$

So $OC = 5OP$.

1 mark
Compare coefficients.

OC and OP are parallel and share the point O.

Therefore O, P and C lie on one straight line.

1 mark
A scalar multiple with a common point proves collinearity.

(Total for Question 20 is 5 marks)

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